

Longqing Zhao

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EDUCATION BACKGROUND

PhD. in School of Water Resources and Hydropower Engineering, Wuhan University

Supervisors: Prof. Jiyun Song

2025-Now

Master in School of Water Resources and Hydropower Engineering, Wuhan University

Supervisors: Prof. Xueshan Ai

2023-2025

Bachelor in Hydrology and Water Resources, Northwest A&F University

Supervisors: Prof. Baowen Yan

2019-2023

RESEARCH INTERESTS

1. High spatiotemporal resolution urban rainfall nowcasting.
2. Urban inundation risk analysis.
3. Flooding Risks in Wuhan City under Future Land Use Change Scenarios.

PUBLICATIONS

Published Journal Papers and Patents:

1. **ZHAO L**, et al. Short-term Inflow Forecasting Technology and Application for Cascade Reservoirs. *Journal of China Hydrology*, 2025, 45(03):9-16. DOI:10.19797/j.cnki.1000-0852.20240272.
2. **AI X**, **ZHAO L** A device for determining the water level height of a stepped reservoir 202411294834.8[P]. 2024-12-17.

Manuscripts Submitted / Under Review:

1. **ZHAO L**, Song J (2026), Optimizing threshold-aware deep learning losses for extreme rainfall nowcasting: A cross-architecture benchmark. *to be submitted to Journal of Hydrology*.

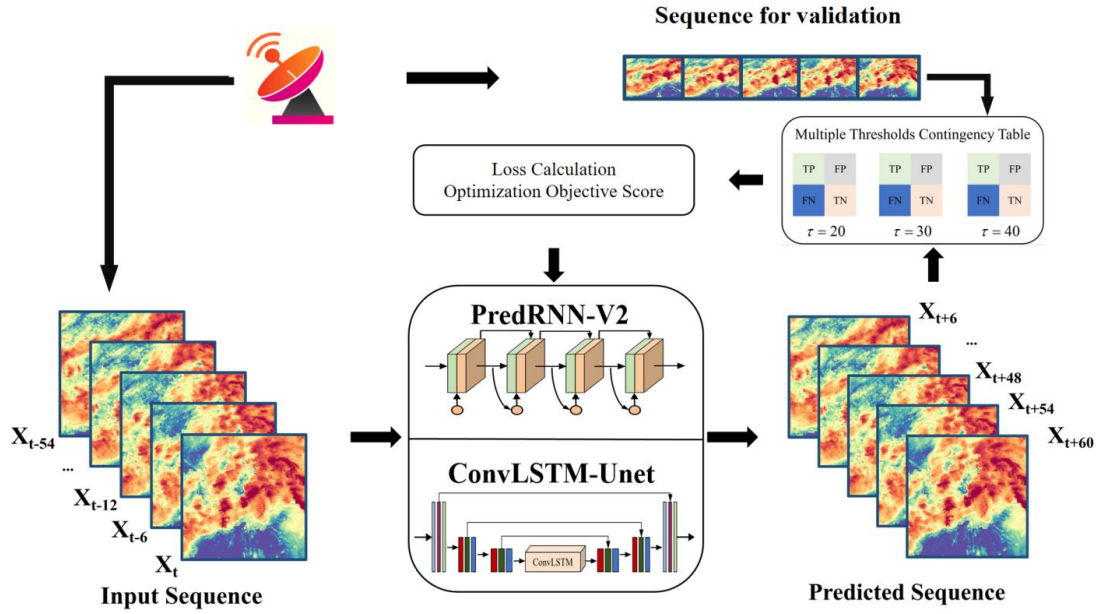
INTERNATIONAL CONFERENCE

1. **ZHAO L**, Song J (2026), Chained Nowcasting of High Spatiotemporal Resolution Urban Rainfall and Inundation Risk Using Deep Learning. *European Geosciences Union 2026 (EGU)*, May 2026, Vienna, Austria. <https://doi.org/10.5194/egusphere-egu26-11991>
2. **ZHAO L**, **AI X** (2024), Research on Short-term Inflow Forecasting Method for Cascade Hydropower Stations in Data-scarce Regions. *The 8th China Africa Water Association (CAWA) Conference (2024)*. July 2024, Nairobi, Kenya.
3. **ZHAO L**, Gao S (2024), Research on Re-development And Utilization Of Hydropower under the new energy system. *The 8th China Africa Water Association (CAWA) Conference (2024)*. July 2024, Nairobi, Kenya.

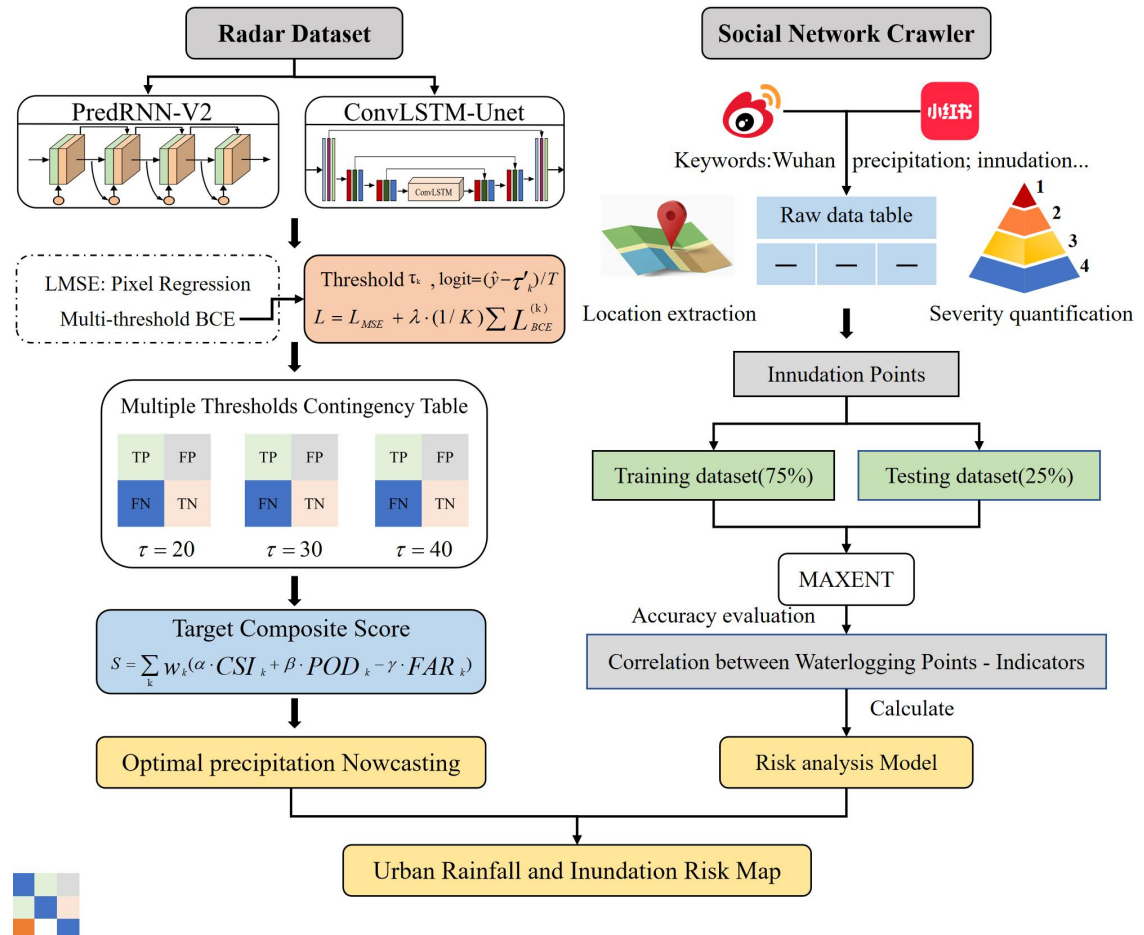
RESEARCH PROJECTS

1. **Radar-based extreme rainfall nowcasting**: improving its spatiotemporal accuracy for heavy

rainfall events through deep learning.



2. Urban Inundation Risk Analysis: coupling the nowcasted rainfall with distributed hydrological-hydraulic models to simulate flood processes and assess inundation risks, supporting emergency planning and real-time decision-making.



RESEARCH EXPERIENCE

- 1. Deep Learning for Heavy-Rain Radar Nowcasting [Prof. Jiyun Song]** 2025-Now
Spatiotemporal deep learning Generative nowcasting
Proposed CT-BCE multi-threshold joint loss with heterogeneous implementations on PredRNN-V2 and ConvLSTM-UNet; designed loss-only benchmarks and manuscript (in preparation).
Built a unified bin.bz2 radar platform adapting PredRNN-V2, ConvLSTM-UNet, NowcastNet, and DGMR with per-lead CSI/FSS evaluation and Slurm HPC training.
Designed Pivot-Frontier (frontier scoring, sparse pivots, bounded refinement, advection-residual decoding) and integrated it with the shared radar data pipeline.
- 2. Research on Flooding Risks in Wuhan City under Future Land Use Change Scenarios** 2025
- 3. Water-heat carbon exchange and environmental response mechanisms in Wuhan City** 2024
- 4. Research on Design Water Levels for Dikes in the Dongting Lake Region and Activation Plans for Flood Storage Areas under the New Flood Control Framework** 2024
- 4. Research on Key Technologies for Multi-dimensional Compensation Dispatch of Hongshui River Cascade Power Stations under High Proportion of New Energy** 2023-2024
- 5. Study on the Variation Law of Water Consumption Rate of Cascade Power Stations under Different Conditions** 2023-2024

TECHNICAL SKILLS

Languages & frameworks: Python; PyTorch (custom training, multi-GPU); PyTorch Lightning; NumPy; experiment config (YAML/argparse)

Deep learning: spatiotemporal forecasting (PredRNN/PredRNN-V2, ST-LSTM, ConvLSTM, ConvGRU); U-Net & attention; hybrid PredRNN-UNet architectures; generative nowcasting (GAN-based NowcastNet, DGMR-style radar generators)

Loss & optimization: regression (MSE) + multi-threshold binary classification (CT-BCE); class imbalance handling (pos_weight, temperature-scaled logits)

Domain: radar echo nowcasting (0–2 h, 5-min lead times); CMA bin.bz2 via cinrad; dBZ normalization; operational metrics (CSI, POD, FAR, FSS) with per-lead evaluation

Data & geospatial: frame caching; ROI cropping with shapefile/shapely and rasterio/GeoTIFF export

MLOps / HPC: Slurm job scripts; early stopping on business scores; unified benchmarking across four SOTA backbones

Research: architecture design (Pivot-Frontier: flow propagation, sparse pivot refinement, advection-residual decoding); academic writing on loss–metric alignment in radar nowcasting

Language: Chinese (native), English CET-4 (554), CET-6 (430)